Final exam study guide

How to prepare — cumulative (Units I–III)

Last updated: October 12, 2026.

Exam: Final exam · Wednesday, December 16, 10:20 a.m.–12:10 p.m., FR 201. Practice problems: Practice packet. Schedule: Schedule.

Recommended study plan

1. **Scan the learning outcomes below** (and the linked midterm study guides). Mark each as: confident / shaky / blank.
2. **Rework Unit I and Unit II with the midterm packets.** Use Midterm 1 study guide / practice and Midterm 2 study guide / practice for fluency you have already practiced once.
3. **Go through Lectures 35–40** for material since Midterm 2 (eigenvectors, complex numbers, least squares, SVD concepts). Match shaky outcomes to fill-ins, worked examples, and Check yourself items.
4. **Build a cumulative "important definitions and formulas" list.** Include vectors and geometry, matrices / Ax / systems, dynamics, eigenpairs, least-squares residual language, and SVD rank-k ideas — not only the newest topics.
5. **Rework quiz styles** from the whole term (Quizzes 1–11). Cover solutions, redo by hand, then check filled-in notes in Canvas.
6. **Start memorizing your list.**
7. **Try the final practice packet with no notes** — Sections A–C by unit, then Section D mixed.
8. **Check answers** (end of the practice packet). For every miss, update your list and redo a similar lecture or midterm-practice item.
9. **Memorize the final list** so you can write it from memory before the exam. In class you will not have notes.

Useful course materials

Source	Role
Lectures 01–40	Full course (skip midterm / Veterans Day / fall-break days as noted on the schedule)
Midterm 1 study guide + practice	Unit I refresh (Lectures 01–14)
Midterm 2 study guide + practice	Unit II refresh (Lectures 18–32)
Quizzes 1–11	Fluency checkpoints across the term
Final practice packet	Cumulative practice (Units I–III + mixed)
Boyd & Vandenberghe, *VMLS*	Textbook (free online)
Strang §6.1	Eigenvalues / characteristic equation
Lecture 40	SVD and low-rank context (conceptual fluency on the final)

Unit I — Vectors and geometry (Lectures 01–14)

Use the Midterm 1 study guide for the full outcome list. High-priority skills for the final:

1. Represent data as vectors; add, scale, and form linear combinations.
2. Compute and interpret inner products, norms, distances, RMS, angles, and cosine similarity.
3. Distinguish linear vs affine models; evaluate $a^{\mathsf{T}}x + b$ in context.
4. Run a small nearest-neighbor or k-means step by hand and discuss scale.
5. Use dependence, span, subspace, basis, dimension, and orthonormal language accurately.

Unit II — Matrices, systems, and dynamics (Lectures 18–32)

Use the Midterm 2 study guide for the full outcome list. High-priority skills for the final:

1. Compute Ax, interpret columns, and recognize common 2D transforms.
2. Translate contexts into $Ax = b$; solve small systems; read RREF (pivots, free variables, consistency).
3. Model $x_{t+1} = Ax_t$; compute a few steps; relate $A^k x$ to long-run behavior.
4. Multiply matrices; compose transforms; compare computational orders.
5. Use invertibility, rank, column space, null space, and 2×2 / triangular determinants.

Unit III — Since Midterm 2 (Lectures 35–40)

Lecture 35 — Eigenvectors as persistent directions

1. Explain geometrically what makes v an eigenvector.
2. Verify a proposed eigenpair (A, v, λ) by checking $Av = \lambda v$.
3. Find λ when an eigenvector is given.
4. Connect eigenvectors to repeated transformations and steady directions.
5. Recognize that a rotation may have no real eigenvector, motivating complex numbers.

Lecture 36 — Characteristic equation, 2×2 eigenproblems, and complex numbers

1. Interpret $a + bi$ algebraically and as a point/vector in the complex plane.
2. Perform the basic complex arithmetic needed for eigenvalue calculations.
3. Form and solve $\det(A - \lambda I) = 0$ for a 2×2 matrix.
4. Find corresponding eigenvectors (including simple complex cases at the course's level).
5. Check real or complex eigenpairs by substitution.

Lecture 37 — Why did the system behave that way?

1. Compute eigenvalues and eigenvectors with software and verify $Av \approx \lambda v$.
2. Use dominant eigenbehavior to explain growth, decay, or approach to a direction.
3. Interpret complex eigenvalues as rotation / oscillation behavior.
4. Observe orthogonal eigenvectors for a symmetric example when it appears.
5. Connect eigen-language back to earlier dynamical simulations.

Lecture 38 — When $Ax = b$ has no exact solution

1. Recognize an inconsistent overdetermined system.
2. Formulate the least-squares objective $\min_x \|Ax - b\|$.
3. Interpret the fitted vector $\hat{b} = A\hat{x}$ as the closest attainable vector in $C(A)$.
4. Interpret the residual $r = b - A\hat{x}$ geometrically.
5. Distinguish an exact solution from a best approximate solution.

Lecture 39 — Least squares, data fitting, and positive definiteness

1. Compute a least-squares solution with software (and check a tiny case by hand when asked).
2. Interpret residuals and RMS error in context.
3. Check that the residual is orthogonal to the columns of A (normal equations idea).
4. Recognize that $A^\mathsf{T}A$ is symmetric positive semidefinite, and positive definite when A has independent columns.
5. Evaluate whether a fitted model is useful, not only report coefficients.

Lecture 40 — SVD, low-rank approximation, and capstone launch

1. Interpret $A = U\Sigma V^{\mathsf{T}}$ conceptually (orthogonal factors + singular values).
2. Explain what a rank-k approximation keeps and what it discards.
3. Compare singular-value decay qualitatively for different data matrices.
4. Relate reconstruction error and storage to the choice of rank k.
5. State a practical limitation of a low-rank model (what structure it may miss).

Cumulative connections (what the final emphasizes)

Be ready to **choose a tool** and **explain why**, not only compute in isolation:

Situation	Tools that often help
Compare or cluster feature vectors	Norm, distance, angle / cosine, nearest neighbor, k-means
"Is b attainable exactly?"	Column space, RREF consistency, rank
"All solutions / free variables"	RREF, null space
"Undo a reversible transform"	Inverse / unique solution of $Ax = b$
"Where is the state after many steps?"	$A^k x$, eigenvalues / dominant direction
"Best approximate fit when overdetermined"	Least squares, residual, RMS
"Compress / approximate a matrix or image"	SVD / rank-k idea
"Hand vs software"	Small 2×2 / 3×3 by hand; larger or SVD/eigen via software interpretation

Capstone studio and gallery (Lectures 41–42) deepen SVD practice but are **not** new quiz days; Friday Dec 11 is **cumulative review only** (no new quiz).